

TET1 displays catalytic and non-catalytic functions in the adult mouse cortex.

PMID: 38970823 | Indexed in PubMed

TET1/2/3 dioxygenases iteratively demethylate 5-methylcytosine, beginning with the formation of 5-hydroxymethylcytosine (5hmC). The post-mitotic brain maintains higher levels of 5hmC than most peripheral tissues, and TET1 ablation studies have underscored the critical role of TET1 in brain physiology. However, deletion of Tet1 precludes the disentangling of the catalytic and non-catalytic functions of TET1. Here, we dissect these functions of TET1 by comparing adult cortex of Tet1 wildtype (Tet1 WT), a novel Tet1 catalytically dead mutant (Tet1 HxD), and Tet1 knockout (Tet1 KO) mice. Using DNA methylation array, we uncover that Tet1 HxD and KO mutations perturb the methylation status of distinct subsets of CpG sites. Gene ontology (GO) analysis on specific differential 5hmC regions indicates that TET1's catalytic activity is linked to neuronal-specific functions. RNA-Seq further shows that Tet1 mutations predominantly impact the genes that are associated with alternative splicing. Lastly, we performed High-performance Liquid Chromatography Mass-Spectrometry lipidomics on WT and mutant cortices and uncover accumulation of lysophospholipids lysophosphatidylethanolamine and lysophosphatidylcholine in Tet1 HxD cortex. In summary, we show that Tet1 HxD does not completely phenocopy Tet1 KO, providing evidence that TET1 modulates distinct cortical functions through its catalytic and non-catalytic roles.

The emerging role of TET enzymes in the immune microenvironment at the maternal-fetal interface during decidualization and early pregnancy.

PMID: 36685517 | Indexed in PubMed

A dysregulated immune microenvironment at the maternal-fetal interface in early pregnancy may lead to early pregnancy loss, fetal growth restriction, and preeclampsia. However, major questions about how epigenetic modifications regulate the immune microenvironment during the decidualization process and embryo implantation remain unanswered. DNA methylation, the main epigenetic mechanism involved in the endometrial cycle, is crucial for specific transcriptional networks associated with endometrial stromal cell (ESC) proliferation, hormone response, decidualization, and embryo implantation. Ten-eleven translocation (TET) enzymes, responsible for catalyzing the conversion of 5-methylcytosine to 5-hydroxymethylcytosine, 5-formylcytosine, and 5-carboxylcytosine to achieve the DNA demethylation process, appear to play a critical role in decidualization and embryo implantation. Here, we provide a comprehensive view of their structural similarities and the common mechanism of regulation in the microenvironment at the maternal-fetal interface during decidualization and early pregnancy. We also discuss their physiological role in the decidual immune microenvironment. Finally, we propose a key hypothesis regarding TET enzymes at the maternal-fetal interface between decidual immune cells and ESCs. Future work is needed to elucidate their functional role and examine therapeutic strategies targeting these enzymes in pregnancy-related disease preclinical models, which would be of great value for future implications in disease diagnosis or treatment.

The N-terminal domain of TET1 promotes the formation of dense chromatin regions refractory to transcription.

PMID: 35235010 | Indexed in PubMed

TET (ten-eleven translocation) enzymes initiate active cytosine demethylation via the oxidation of 5-methylcytosine. TET1 is composed of a C-terminal domain, which bears the catalytic activity of the enzyme, and a N-terminal region that is less well characterized except for the CXXC domain responsible for the targeting to CpG islands. While cytosine demethylation induced by TET1 promotes transcription, this protein also interacts with chromatin-regulating factors that rather silence this process, the coordination between these two opposite functions of TET1 being unclear. In the present work, we uncover a new function of the N-terminal part of the TET1 protein in the regulation of the chromatin architecture. This domain of the protein promotes the establishment of a compact chromatin architecture displaying reduced exchange rate of core histones and partial dissociation of the histone linker. This chromatin reorganization process, which does not rely on the CXXC domain, is associated with a global shutdown of transcription and an increase in heterochromatin-associated histone epigenetic marks. Based on these findings, we propose that the dense chromatin organization generated by the N-terminal domain of TET1 could contribute to restraining the transcription enhancement induced by the DNA demethylation activity of this enzyme.

CRISPR/Cas9-mediated demethylation of FOXP3-TSDR toward Treg-characteristic programming of Jurkat T cells.

PMID: 34954490 | Indexed in PubMed

Demethylation of FOXP3-TSDR (Treg specific demethylated region) is a hallmark of stable differentiation and suppressive function of regulatory T (Treg) cells. Previous protocols aiming at human naïve T cell differentiation failed to implement a Treg cell specific epigenetic signature. Ten-eleven translocation (TET) enzymes catalyze DNA demethylation. Plasmids toward expression of a fusion protein encompassing nonfunctional Cas9, the catalytic domain of TET1, blue fluorescent protein, and encoding single guide RNAs (sgRNAs) targeting specific segments of the FOXP3-TSDR were engineered and transfected into Jurkat T cells. FOXP3-TSDR methylation was analyzed by deep-amplicon bisulfite sequencing while cellular Foxp3, Tbet, Gata3, and Ror γ t mRNA levels were determined by real-time PCR. Overexpression of dCas9-TET1 significantly decreased Jurkat cell FOXP3-TSDR methylation and increased Foxp3 mRNA expression while expressions of master transcription factor mRNAs of other major T cell lineages remained largely unaffected. dCas9-TET1 construct transfection mediated Treg programming of patients' primary T cells might be feasible.

Ten-eleven translocase: key regulator of the methylation landscape in cancer.

PMID: 33913031 | Indexed in PubMed

Methylation of 5th residue of cytosine in CpG island forms 5-methylcytosine which is stable, heritable epigenetic mark. Methylation levels are broadly governed by methyltransferases and demethylases. An aberration in the demethylation process contributes to the silencing of gene expression. Ten eleven translocation (TET) dioxygenase (1-3) the de novo demethylase is responsible for conversion of 5-methylcytosine (5-mC) to 5-hydroxymethylcytosine (5-hmC), 5-formylcytosine (5-fC) and 5-carboxycytosine (5-caC) during demethylation process. Mutations and abnormal expression of TET proteins contribute to carcinogenesis. Discovery of TET proteins has offered various pathways for the

reversal of methylation levels thus, enhancing our knowledge as to how methylation effects cancer progression. We searched "PubMed" and "Google scholar" databases and selected studies with the following keywords "TET enzyme", "cancer", "5-hmC", and "DNA demethylation". In this review, we have discussed combinatorial use of vitamin C in inhibiting tumour growth by enhancing the catalytic activity of TET enzymes and consequently, increasing the 5-hmC levels. 5-Hydroxymethylcytosine holds promise as a prognostic biomarker in solid cancers. The contribution of induction and suppression of TET enzymes and 5-hmC carcinogenesis are discussed in haematological and solid cancers. We found that TET enzymes play central role in maintaining the methylation balance. Any anomaly in their expression may dip the balance towards cancer progression. Low levels of TET enzymes and 5-hmC correlate with tumour invasion, progression and metastasis. Also, use of vitamin C enhances TET activity. TET enzymes play vital role in shaping the methylation landscape in body. 5-hmC can be used as prognostic marker in solid cancers.

The roles of TET family proteins in development and stem cells.

PMID: 31941705 | Indexed in PubMed

Ten-eleven translocation (TET) methylcytosine dioxygenases are enzymes that catalyze the demethylation of 5-methylcytosine on DNA. Through global and site-specific demethylation, they regulate cell fate decisions during development and in embryonic stem cells by maintaining pluripotency or by regulating differentiation. In this Primer, we provide an updated overview of TET functions in development and stem cells. We discuss the catalytic and non-catalytic activities of TETs, and their roles as epigenetic regulators of both DNA and RNA hydroxymethylation, highlighting how TET proteins function in regulating gene expression at both the transcriptional and post-transcriptional levels.

A twist in zygotic reprogramming.

PMID: 26820436 | Indexed in PubMed

The first hours of mammalian embryogenesis are devoted to extensive epigenetic reprogramming. One hallmark is active demethylation of the paternal genome by Tet (ten-eleven translocation) enzymes. However, the process is now shown to be Tet-independent at first, with Tet enzymes only counteracting hitherto underappreciated de novo DNA methylation activity in later zygotic stages.

TET proteins in cancer: Current 'state of the art'.

PMID: 26276226 | Indexed in PubMed

Aberrations in DNA methylation patterns are observed from the early stages of carcinogenesis. However, the mechanisms that drive these changes remain elusive. The recent characterization of ten-eleven translocation (TET) enzymes as a source of newly modified cytosines (5-hydroxymethylcytosine, 5-formylcytosine and 5-carboxylcytosine) has shed new light on the DNA demethylation process. These cytosines are intermediates of an active DNA demethylation process and are epigenetic markers per se. In this review, we discuss the mechanism and function of TET proteins in biological processes as well as

current knowledge regarding their expression and regulation in cancer.

Drosophila Tet Is Required for Maintaining Glial Homeostasis in Developing and Adult Fly Brains.

PMID: 35396259 | Indexed in PubMed

Ten-eleven translocation (TET) proteins are crucial epigenetic regulators highly conserved in multicellular organisms. TETs' enzymatic function in demethylating 5-methyl cytosine in DNA is required for proper development and TETs are frequently mutated in cancer. Recently, *Drosophila melanogaster* Tet (dTet) was shown to be highly expressed in developing fly brains and discovered to play an important role in brain and muscle development as well as fly behavior. Furthermore, dTet was shown to have different substrate specificity compared with mammals. However, the exact role dTet plays in glial cells and how ectopic TET expression in glial cells contributes to tumorigenesis and glioma is still not clear. Here, we report a novel role for dTet specifically in glial cell organization and number. We show that loss of dTet affects the organization of a specific glia population in the optic lobe, the "optic chiasm" glia. Additionally, we find irregularities in axon patterns in the ventral nerve cord (VNC) both, in the midline and longitudinal axons. These morphologic glia and axonal defects were accompanied by locomotor defects in developing larvae escalating to immobility in adult flies. Furthermore, glia homeostasis was disturbed in dTet-deficient brains manifesting in gain of glial cell numbers and increased proliferation. Finally, we establish a *Drosophila* model to understand the impact of human TET3 in glia and find that ectopic expression of hTET3 in dTet-expressing cells causes glia expansion in larval brains and affects sleep/rest behavior and the circadian clock in adult flies.

TET Family of Dioxygenases: Crucial Roles and Underlying Mechanisms.

PMID: 26302812 | Indexed in PubMed

DNA methylation plays an important role in the epigenetic regulation of mammalian gene expression. TET (ten-eleven translocation) proteins, newly discovered demethylases, have sparked great interest since their discovery. TET proteins catalyze 5-methylcytosine to 5-hydroxymethylcytosine, 5-formylcytosine and 5-carboxylcytosine in 3 consecutive Fe(II)- and 2-oxoglutarate (2-OG)-dependent oxidation reactions. TET proteins dynamically regulate global or locus-specific 5-methylcytosine and/or 5-hydroxymethylcytosine levels by facilitating active DNA demethylation. In fact, in addition to their role as methylcytosine dioxygenases, TET proteins are closely related to histone modification, interact with metabolic enzymes as well as other proteins, and cooperate in transcriptional regulation. In this review, we summarize the recent progress in this exciting field, highlighting the molecular mechanism by which TET enzymes regulate gene expression and their functions in health and disease. We also discuss the therapeutic potential of targeting TET proteins and aberrant DNA modifications.
